

SAGAR SONAWANE

AI/ML ENGINEER (DATA SCIENTIST)

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SUMMARY

AI/ML Engineer with 4+ years of experience in developing scalable AI and ML solutions in healthcare and finance domains. Expertise in speech-to-text systems, object detection, NLP, and GenAI models. Skilled in Python, machine learning frameworks, and cloud technologies with a strong background in statistical analysis, model optimization, and deployment.

TECHNICAL SKILLS

Programming & Tools: Python, Flask, OpenCV, LangChain, TensorFlow, **Databases:** SQL, VectorDB (Pinecone)
Other Tools: Git, JIRA, Streamlit, Elasticsearch, postman

Machine Learning: Regression models, SVM, Decision Trees, Random Forest, XGBoost, K-Means.
Deep Learning: CNN, RNN, ANN, LSTM, Transfer Learning.

NLP: NLTK, Spacy, BERT, Word2Vec, TF-IDF, Hugging Face ,Gemini, LLM, NVIDIA Nemo.
Cloud & DevOps: AWS (EC2, S3, SageMaker), Docker

PROFESSIONAL EXPERIENCE

Ai ML Engineer , State Bank of India- Software Factory

Oct 2024 - Present

- Developed CRCF chatbot for SBI using BERT for natural language understanding and Elasticsearch for efficient document retrieval, enhancing customer service automation.

AI Engineer, Middletown Medical, New York, USA

JUN 2024 - October 2024

- Implemented a speech-to-text conversion system using Nvidia Nemo and Whisper for accurate transcription of doctor-patient conversations. Also build Fall detection for patient monitoring
- Optimized models for medical terminology, improved audio quality, and integrated preprocessing techniques to handle noisy data.
- Developed batch processing pipelines and collaborated with medical staff to validate transcriptions.

ML Developer | 3.5 Years | Nuance, Pune, India

JAN 2021 - MAY 2024

- Built a credit card defaulter prediction model using logistic regression, Naïve Bayes, and XGBoost for customer credit risk analysis.
- Build chatbot Development and Real-time Object detection
- Performed data preprocessing, clustering with K-Means, and hyperparameter tuning using Search for improved model performance. Deployed APIs with Flask for predictive analysis results.

PROJECTS

Speech-to-Text Conversion

- Developed a speech-to-text model leveraging Nvidia Nemo, Whisper, and Deepgram APIs to improve transcription accuracy in noisy environments.

Chatbot Development

- Built a domain-specific chatbot using fine-tuned BERT and Gemini models, integrated with a RAG pipeline for context-aware interactions.

Real-Time Person Detection

- Designed a real-time detection system using YOLOv4 and YOLOv7 in healthcare settings for patient monitoring and security.

Credit Card Defaulter Prediction

- Implemented clustering and classification models to assess credit repayment likelihood, improving financial decision-making for clients.

CRCF Chatbot for State Bank of India (SBI)

Tech Stack: BERT, Elasticsearch, Python, Flask , NLP, REST APIs, Postman

- Built CRCF (Customer Redressal & Complaint Filing) chatbot for SBI using BERT for intent detection and Elasticsearch for fast and accurate query retrieval, improving automation of customer complaint handling.

ADDITIONAL INFORMATION

- **Education:** BE in Electronics & Telecommunication Engineering | DKTE Society's Textile & Engineering Institute, Ichalkaranji | First Class.
- **Certifications:** AWS Certified Solutions Architect, Advanced Machine Learning Specialization (Coursera)